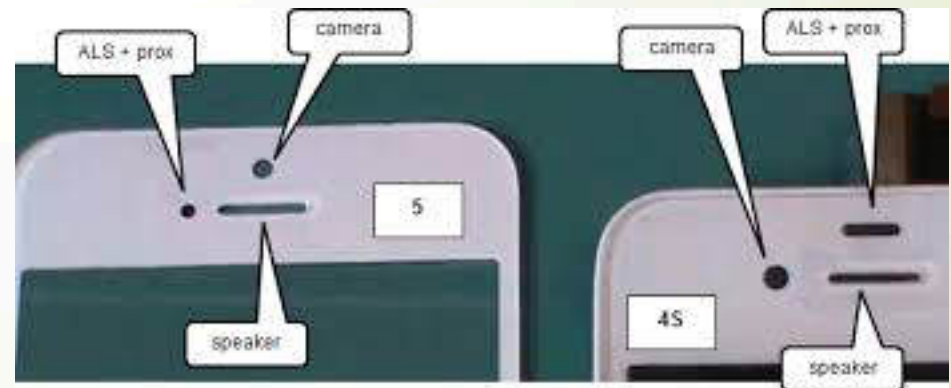


Examples

- Phone display adjusts the brightness of the display based on the surrounding area
- Uses a light sensor



Examples

- Device displays user's location, shows route to a desired destination, find nearby stores, geotag images on social media, etc.
- Uses location sensor



User-Interface leveraging Contextual (Location) Information

Name	Room	Distance
caps	35-2200	200ft
claudia	35-2108	30ft
perfector	35-2301	20ft
snoball	35-2103	100ft

(a)

Distance	Name	Room
20ft	perfector	35-2301
30ft	claudia	35-2108
100ft	snoball	35-2103
200ft	caps	35-2200

(b)

Name	Room	Distance
caps	35-2200	200ft
claudia	35-2108	30ft
perfector	35-2301	20ft
snoball	35-2103	100ft

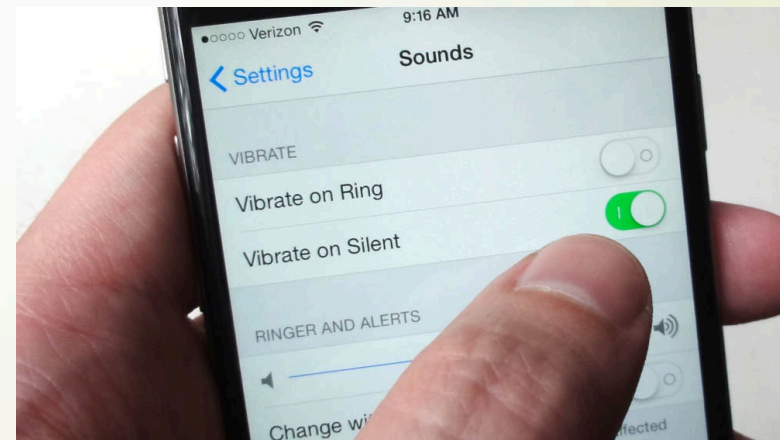
(c)

Name	Room	Distance
caps	35-2200	200ft
claudia	35-2108	30ft
perfector	35-2301	20ft
snoball	35-2103	100ft

(d)

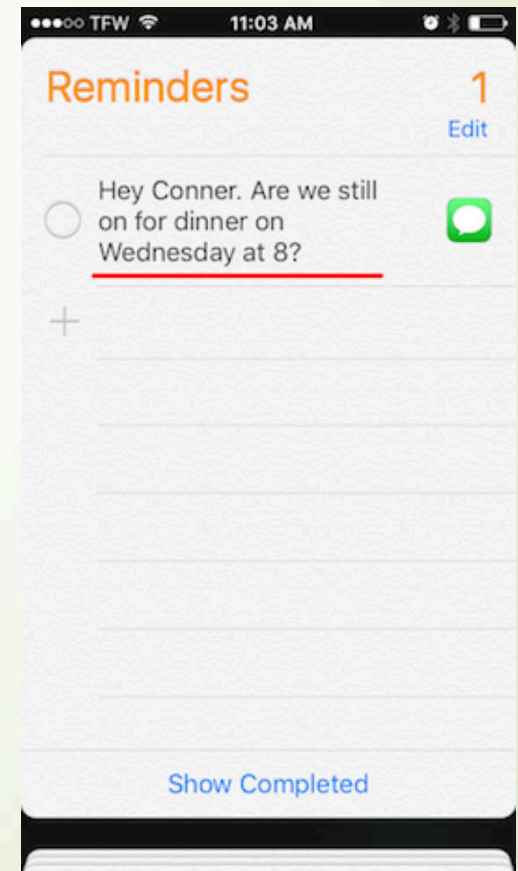
Automatic Contextual Reconfiguration

- Add, remove, or alter components based on context
- Smart notifications on phone (ring, vibrate, auto-response)
- Could be largely based on sensor readings (may discuss more next time)



Context-Triggered Actions

- Simple if-then condition-action rules, automatically invoked
- Reminder: if I step into the car on weekday morning and don't have suitcase with me, remind me to get it
- Although tremendous advances have been made, context-aware systems still haven't freed us from manually creating memos/schedules!

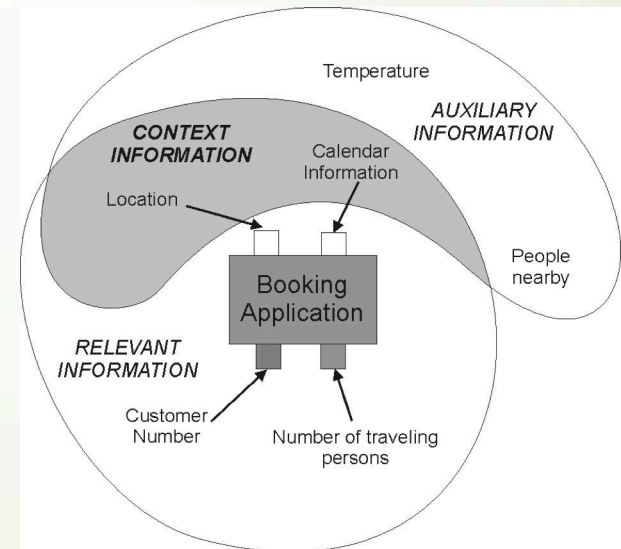


Why Use Context?

- **Reduce cognitive load of user**
- **Proactivity**
 - Set up environment according to user's preferences/history
 - Auto-completion of forms (location, time in timetable)
 - Reminders
- **Search and filter information** according to user's needs
- **Avoid interrupting** the user in inappropriate situations
- **Smart environments**
 - Turn devices on/off, start applications, ... depending on location, time, situation (lecture, meeting, home cinema, ...)
 - Discover and use nearby interaction devices

Definitions of Context

- ▶ “Context is **any information that can be used to characterize the situation of an entity**. An entity is a person, place, or object that is considered **relevant** to the interaction between a user and an application, including the user and applications themselves” [Dey et al. 2001]
- ▶ Auxiliary: not essential
- ▶ Relevant: can actually be used



Classification

➤ External (physical)

- Context that can be measured by hardware sensors
- Examples: location, light, sound, movement, touch, temperature, air pressure, etc.

➤ Internal (logical)

- Mostly specified by the user or captured monitoring the user's interaction
- Examples: the user's goal, tasks, work context, business processes, the user's emotional state, etc.

Agenda

- Location-Based Services
- Context-Aware Systems
- Challenges and Design Guidelines
- Sample Work: Coordinated Sensor Activations

Challenges: Self-Awareness

- Context-awareness helps technology to “get it right”
- But context is hard to sense (quantity, subtleness)
- Computers are not self-aware like humans
- When the system does the wrong thing
 - auto-locking car doors
 - screen saver during presentation
 - microphone amplifying a whisper

Challenges: Intelligence

- ▶ Context data must be coupled with the ability to interpret it, but computers are bad at “common sense”.
- ▶ More rules $\neq$ intelligence
- ▶ More rules = more complexity, harder to understand
- ▶ Keep “**Human in the Loop**”?
 - ▶ computers can detect, aggregate, portray information
 - ▶ allow human users to interpret and act on it
 - ▶ is this a good strategy for all context-aware systems?

Challenges: Programming

- Developers have **little experience with devices that gather the data** (e.g., gyroscopes).
- Data gathered from a sensor **must be interpreted correctly** in order for it to be useful.
- Context comes from various sources and in order for this data to be useful it **must be combined correctly** (i.e., the gyroscope and accelerometer working together to determine orientation).
- The context **changes constantly in real time**.

Challenges: Losing Control?

- **Automation** reduces the amount of work that users have to do
- Users like the idea of a device that completes tasks on their behalf
- However, when users use these devices they feel a **loss of control** if a device has a high level of automation

Challenges: Other

- **Privacy**
 - Should law enforcement be able to access the history of a user?
- **Correctness**
 - Errors fusing data
 - Detection errors
 - Interpretation errors
- **Complexity**
 - Difficult to develop, maintain, understand
 - Reduces accuracy of the application
 - Extra resource consumptions!

Challenges: Other (cont.)

- **User preferences**
 - May not match what the device does!
 - Everyone is different!
 - What is your idea of “nighttime”?
 - What is your idea of “warm”? Or “loud”?
- **Information overload**
 - Can overwhelm the user

Solution Approaches

- Keep an appropriate level of automation (avoid uncertainty)
- Avoid unnecessary interruptions
- Avoid information overload
- Keep an appropriate level of system status visibility
- Account for the impact of Social Context
- Allow for the personalization of individual needs
- Secure the user's privacy

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Motivation

- Increasingly more context-aware applications leverage the rich set of sensors on the smartphones.
- These applications directly control sensors which lead to redundant activations and energy waste



Introduction

- Opportunity for optimization?
 - Different apps may require same contexts
 - Use different sensors to sense same contexts
- Research Problem: How to select the most efficient sensing strategy ?
 - Satisfy all apps' requirements
 - Minimize energy consumption



System Overview

- We proposed an **Optimal Sensor Management (OSM)** middleware
- The OSM middleware sits between apps and the hardware
- OSM middleware :
 - Provides API to connect apps
 - Maintains a database of active requests
 - Determines which sensors should be activated